Power Supplies

In its first volume (Part 3), the FORMANT series describes a high-quality power supply unit designed to meet exacting standards. However, for FORMANT circuits where power supply requirements are not critical—such as the expansion circuits described in this book—the aforementioned unit is unnecessarily complex. This applies, for instance, to setups that do not incorporate VCOs. The following description presents a more cost-effective alternative and also provides instructions for construction and assembly.

+15 V / –15 V Power Supply

The schematic for the +15 V / –15 V power supply is shown in Figure 1. The circuit features no unusual characteristics; it is constructed using the well-known fixed-voltage regulators from the 78xx/79xx series. These regulators offer distinct advantages over conventional Zener diode-based stabilization circuits: current limiting, short-circuit protection, and thermal shutdown. The specific types used—housed in TO-220 packages—are employed, for example, in the external "Parametric Equalizer" (Chapter 3). Figure 3 illustrates the PCB layout and the component placement diagram for the +15 V / –15 V power supply.

+5 V Power Supply

VCOs or other synthesizer circuits housed in additional module enclosures—particularly those utilizing (LS) TTL ICs (e.g., digital sequencers)—require a +5 V power source (preferably adjustable) in addition to the standard +15 V / -15 V supply. For this application, a circuit based on the inexpensive and proven µA723 IC was selected (Figure 4). This adjustable integrated voltage regulator exhibits exceptionally good specifications regarding temperature stability and ripple rejection. The circuit design is identical to the +5 V rail of the power supply unit found in the FORMANT series. Here, too, the output current is "boosted" by means of a power transistor (T1).

The choice of T1 depends on the specific current draw of the circuit(s) or the sizing of the power supply. PNP power transistors in SOT-32 packages can be mounted directly onto the power supply's circuit board, complete with their heat sinks. The pinout for such transistors (e.g., BD 137, BD 139) is shown in Figure 5. However, if the power supply is to be designed for a current output exceeding 0.8 amperes, it is preferable—as is the case in the FORMANT series power supply—to use a 2N3055; this transistor, however, must be mounted separately on its own dedicated heat sink.

The output voltage is adjustable via the Cermet trimmer P1, and the output is short-circuit proof, making the power supply suitable for experimental purposes as well. The circuitry of the internal current-limiting transistors (Pins 2 and 3)—configured in a "foldback" arrangement—ensures that short-circuit currents are limited to approximately 0.4 A. When the foldback threshold (maximum output current) is exceeded, the output voltage drops to a level where LED D6 no longer illuminates.

The specifications for the mains transformer (Tr) and resistor R5 depend on the current draw of the circuits being powered. To ensure reliable operation of IC1, the transformer should provide a secondary voltage of at least 9 V. Using a "metal-can" package for IC1 offers superior temperature stability; in this case, the pin connections indicated in parentheses must be bent to match the pin configuration of the DIL package type.

Figure 6 illustrates the PCB layout and component placement plan for the +5 V power supply. Assembly and calibration are straightforward. This +5 V power supply has proven highly effective when used in conjunction with a "Digital Keyboard Controller" (Chapter 1), where a single 5-core cable (COM, KB-Gate, +15 V, –15 V, Ground) serves as the interconnecting link between the module and the keyboard console housing. If the +5 V power supply unit is mounted directly within the keyboard console housing, the +5 V required to power the LS-TTL ICs can be tapped directly from the +15 V supply line.

One more tip

As is well known, the power supplies in the FORMANT are protected against short circuits and overheating. What would also be a sensible addition is a protection circuit against overvoltage. Why? Well, for one thing, the voltage regulator could fail; for another, there is always the risk—especially when experimenting—of accidentally reversing the supply voltages.

Figure 7 illustrates a +5 V overvoltage protection circuit. Its operation is relatively simple. Under normal operating conditions—with a correct supply voltage of +5 V—the gate current flowing into the thyristor is so low that the device remains in its blocking state. However, in the event of an overvoltage at the input, the gate current rises to a level sufficient to trigger the thyristor, thereby short-circuiting the excessively high load current. This action either blows a fuse or temporarily disables the corresponding voltage regulator IC. To provide a visual indication of this protective function, it is advisable to wire the power supply's indicator LED downstream of the thyristor protection circuit.

The type of thyristor depends on the current draw of the connected circuit. In any case, it is recommended to mount it on a heat sink.

Figures

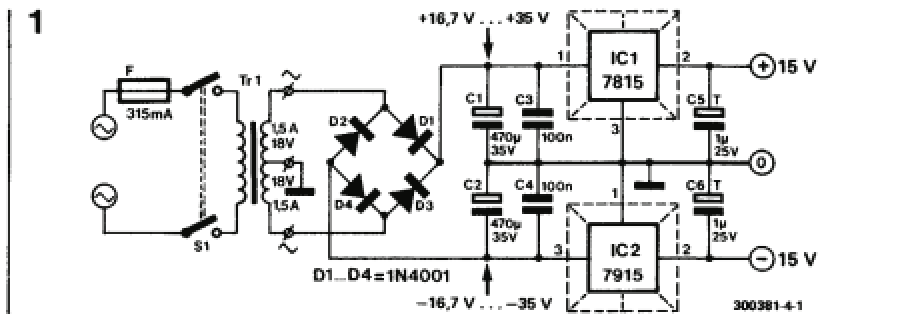

Figure 1. Schematic diagram of the +15 V / -15 V power supply.

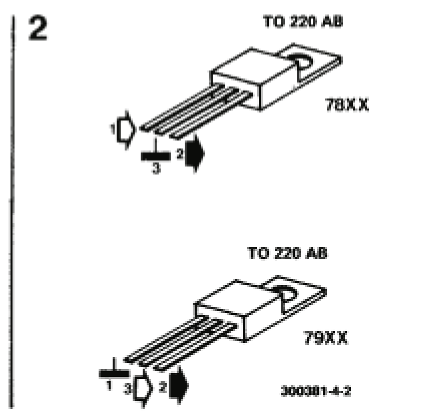

Figure 2. Pinout of the 78xx and 79xx fixed-voltage regulators in a TO-220 package

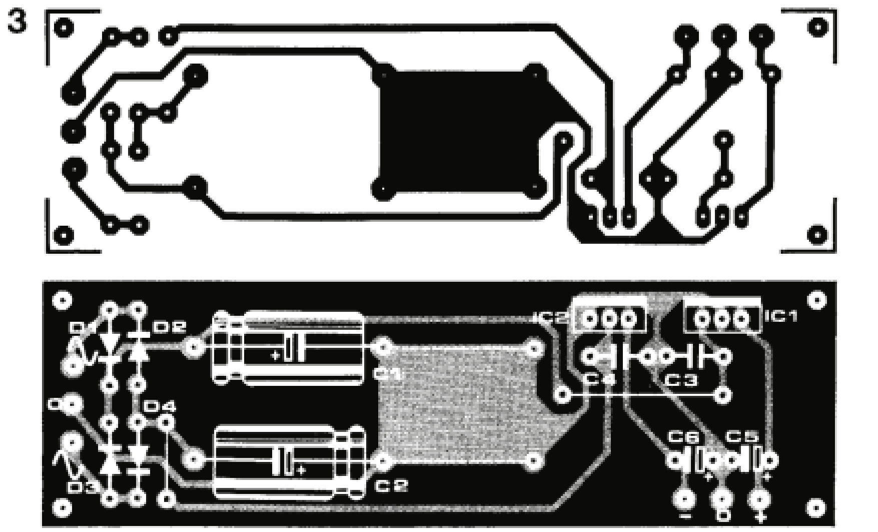

Figure 3. PCB layout and component placement guide for the circuit shown in [Figure 1]

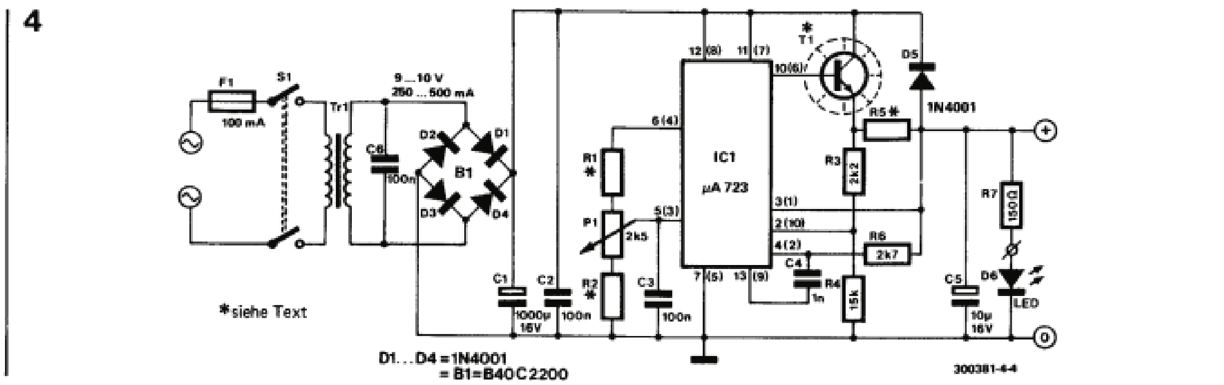

Figure 4. Schematic diagram of the adjustable +5 V power supply.

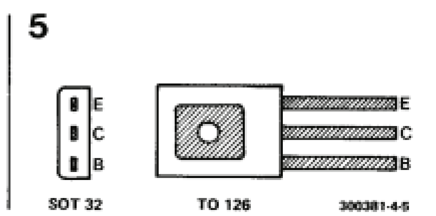

Figure 5. Pinout of NPN transistors (for T1) in a SOT-32 package.

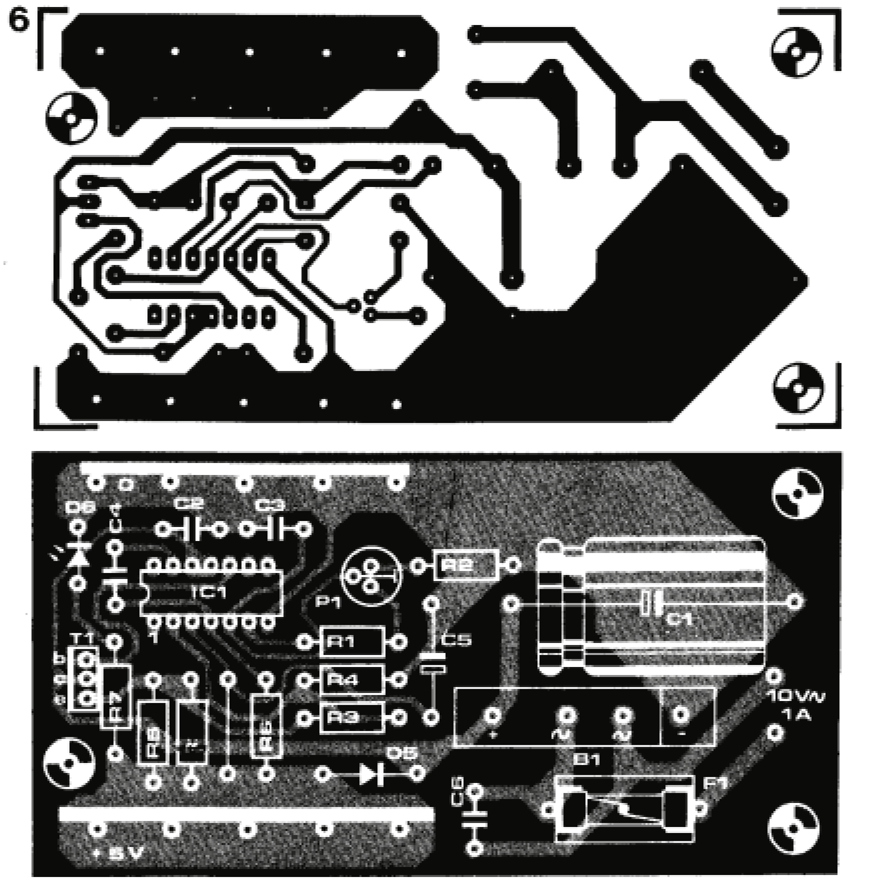

Figure 6. PCB layout and component placement guide for the circuit shown in Figure 4. On the PCB, R5 consists of two resistors connected in parallel

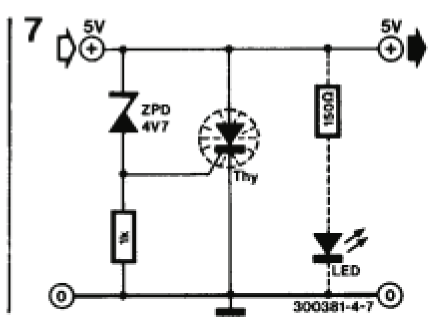

Figure 7. Overvoltage protection circuit

Bill of Materials for Figure 4

Resistors:

- R1 = 2.74k (Metal Film, 1%)
- R2 = 8.25k (Metal Film, 1%)
- R3 = 2.2k (Carbon Film, 5%)
- R4 = 15k (Carbon Film, 5%)
- R5 = 1.5Ω to 3.3Ω / 1W (Carbon Film, 5%)
- R6 = 2.7k (Carbon Film, 5%)
- R7 = 150Ω (Carbon Film, 5%)

Potentiometer:

- P1 = 2.5k (Cermet)

Capacitors:

- C1 = 1000 µF / 16 V
- C2, C3 = 100 nF (MKH, MKS)
- C4 = 1 nF (MKH, MKS)
- C5 = 10 µF / 16 V

Semiconductors:

- D1...D5 = 1N4001
- D1...D4 = B1 = B40C2200
- D6 = LED
- T1 = BD137, BD139, 2N3055 (see text)
- IC1 = µA723, LM723, or similar

Miscellaneous:

- Tr = Transformer, 9 V / 0.5...1.5 A (see text)
- Si = Fuse, 0.1 A
- 1 x 2-pole mains switch

Front Panel Suggestions with Power Switches

The original FORMANT concept does not include a power switch. This poses no problem, provided that the power outlet and plug remain freely accessible. However—particularly in stage settings—the tangle of cables is often so dense that locating the correct power plug requires several failed attempts. With the front panels presented here, along with the accompanying tips, the FORMANT can be switched on and off directly from the front panel.

No dedicated printed circuit board is required for this circuit. The schematic diagram is shown in Figure 1. Unlike the schematic for the power supply unit within the FORMANT series, a notable feature here is the additional 4.7 kΩ resistor used to connect the earth line to ground. When using metal enclosures, it is also advisable to ground the enclosure itself to prevent electromagnetic interference. The schematic shown in Figure 1 applies *mutatis mutandis* to power supply units in expansion enclosures. The power switch and fuse holder are hardwired directly to the front panel. The wire gauge used for these connections should not be undersized (a minimum diameter of 1 mm is recommended).

Figures 2 and 3 illustrate suggested front panel layouts. The wiring for the supply voltage indicator LEDs is shown in Figure 4. When designing a module enclosure for the FORMANT—particularly to accommodate potential expansion circuits—provision for strain relief on the power cord should not be omitted. This point requires particular attention in cases of (semi-)professional use or frequent transport. A cost-effective and practical solution is illustrated in the sketch in Figure 5. A three-conductor power cord with sufficiently robust insulation is recommended for this application.

In the interest of flawless operational safety, please observe the applicable VDE regulations.

Regarding the Mechanical Design

A well-known and recurring problem with DIY electronic devices is the question of finding a suitable enclosure. In some cases, this is precisely where the project fails to reach completion. Either the device remains sitting in a corner as an unfinished torso, or it is put into operation without a housing. The latter violates all safety regulations and also compromises a professional appearance.

The FORMANT is much better off in this regard. The circuit boards typically follow the “Eurocard” format, and the front panels are designed for installation in a 19-inch rack enclosure.

This offers significant advantages during the setup phase:

- Seamless addition of new modules
- High mechanical stability
- Service-friendly design
- Professional appearance

In addition to using profile-based mounting enclosures or wooden cabinets, a combined approach utilizing both wood and aluminum sheets and angles is a viable option. This offers an inexpensive, versatile, and customizable alternative. In this design, the side panels are crafted from natural wood of sufficient thickness (min. 19 mm) or from plastic-coated chipboard. The base, back panel, and top cover plate are made from aluminum sheet metal at least 2 mm thick. Depending on personal preference, the aluminum can be left in its natural silver finish or painted in any other color. Connections between the individual enclosure components are established using 20 x 20 x 2 mm aluminum angles. These angles also serve to mount the front panels. The connections between the individual aluminum support members are best secured using M3 machine screws. Since aluminum is a relatively soft material, cutting the threads should not present any significant difficulties. The material costs for such an enclosure—capable of housing six large and six small FORMANT modules—amount to approximately 60 DM. Incidentally: The FORMANT enclosure can also be customized using Plexiglas.

Bill of Materials for the "Power/Caution" Unit.

Resistors (Carbon Film, 5%):

- 1 x 4.7kΩ

Miscellaneous:

- 1 x Front Panel
- 1 x Miniature Toggle Switch (DPDT, 250 V / 0.5 A)
- 1 x Fuse (0.5 A)
- 1 x Single-Hole Mount Fuse Holder

"Power" Bill of Materials for Expansion Enclosure

Resistors (Carbon Film, 5%):

- 1 x 150 Ω
- 2 x 680 Ω
- 1 x 4.7 kΩ

Semiconductors:

- 3 x LEDs

Miscellaneous:

- 1 x Front Panel
- 1 x Miniature Toggle Switch (DPDT) [250 V / 0.5 A]
- 1 x Fuse (Rated according to current draw)
- 1 x Single-Hole Mount Fuse Holder

Figures

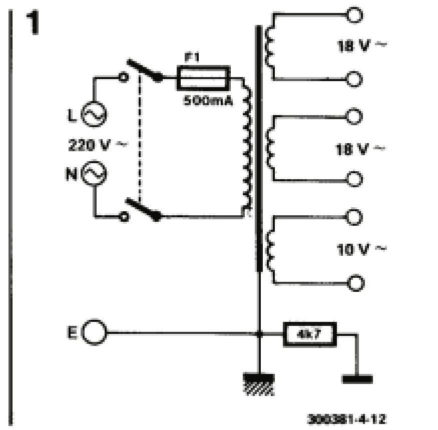

Figure 1. Proposed modification for the FORMANT power supply (Book 1, Part 3).

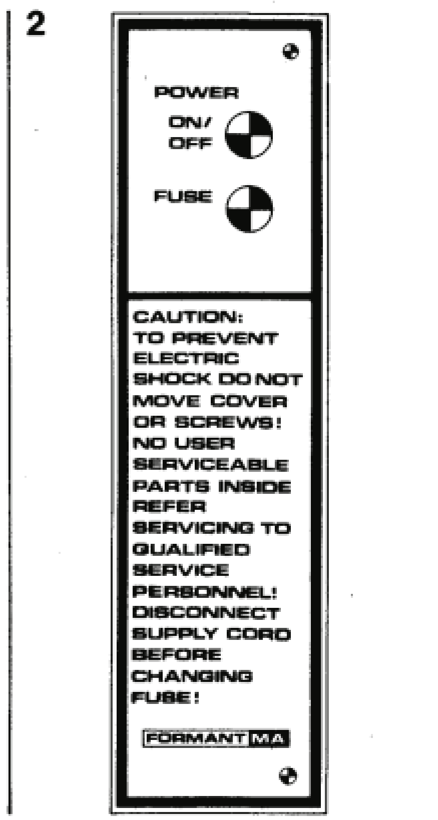

Figure 2. Proposed front panel design for a power switch. This front panel is intended for FORMANT configurations equipped with a COM module. To prevent unauthorized persons from performing improper work on the FORMANT, the warning label reads (in essence): "CAUTION: Do not remove the front panel. Entrust service work to qualified personnel! Disconnect the power plug before replacing the fuse."

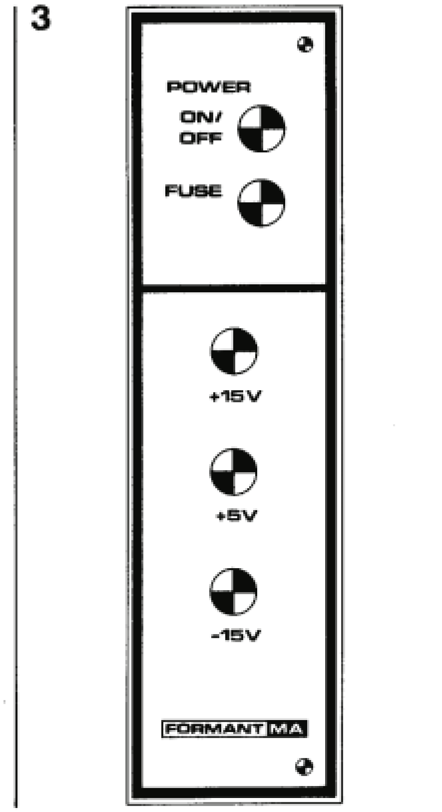

Figure 3. An alternative proposal depicts a power switch front panel for a FORMANT without a COM module. In addition to the power switch, mounting holes are provided for the LEDs indicating the various supply voltages.

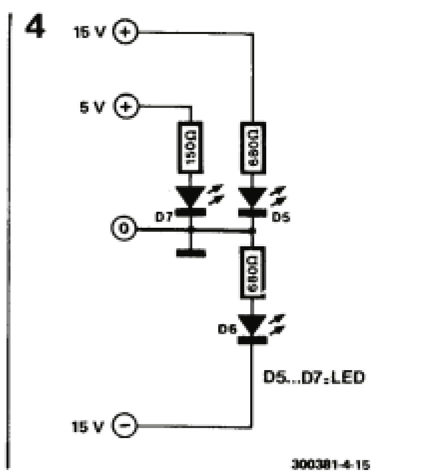

Figure 4. Circuit extension for the visual indication of the various supply voltages. Since expansion module enclosures typically do not require a dedicated interface receiver, no separate "Gate" LED is necessary—unlike the LED indicators found on the COM module.

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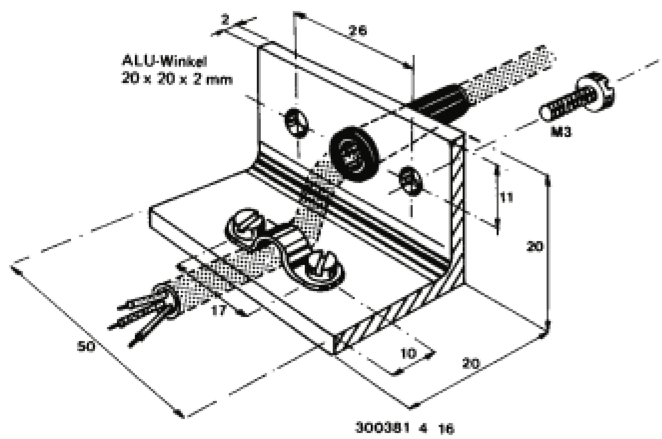

Figure 5. Strain relief for the power cord on the FORMANT module enclosure.